

Problem 2

A helicopter is stationary at a height of $H = 980$ m. A bomb is released from the helicopter and falls freely. After what time will the pilot hear the explosion of the bomb? Assume that the speed of sound is $c = 340$ m/s.

Solution

The time for the bomb to reach the ground is determined using the free-fall equation:

$$H = \frac{1}{2}gt_f^2.$$

Solving for t_f :

$$t_f = \sqrt{\frac{2H}{g}} = \sqrt{\frac{2 \times 980}{9.81}} = \sqrt{199.59} \approx 14.13 \text{ s.}$$

After the explosion, the sound travels back to the pilot. The time for sound to travel is:

$$t_s = \frac{H}{c} = \frac{980}{340} \approx 2.88 \text{ s.}$$

Thus, the total time after which the pilot hears the explosion is:

$$t_{total} = t_f + t_s = 14.13 + 2.88 = 17.01 \text{ s.}$$